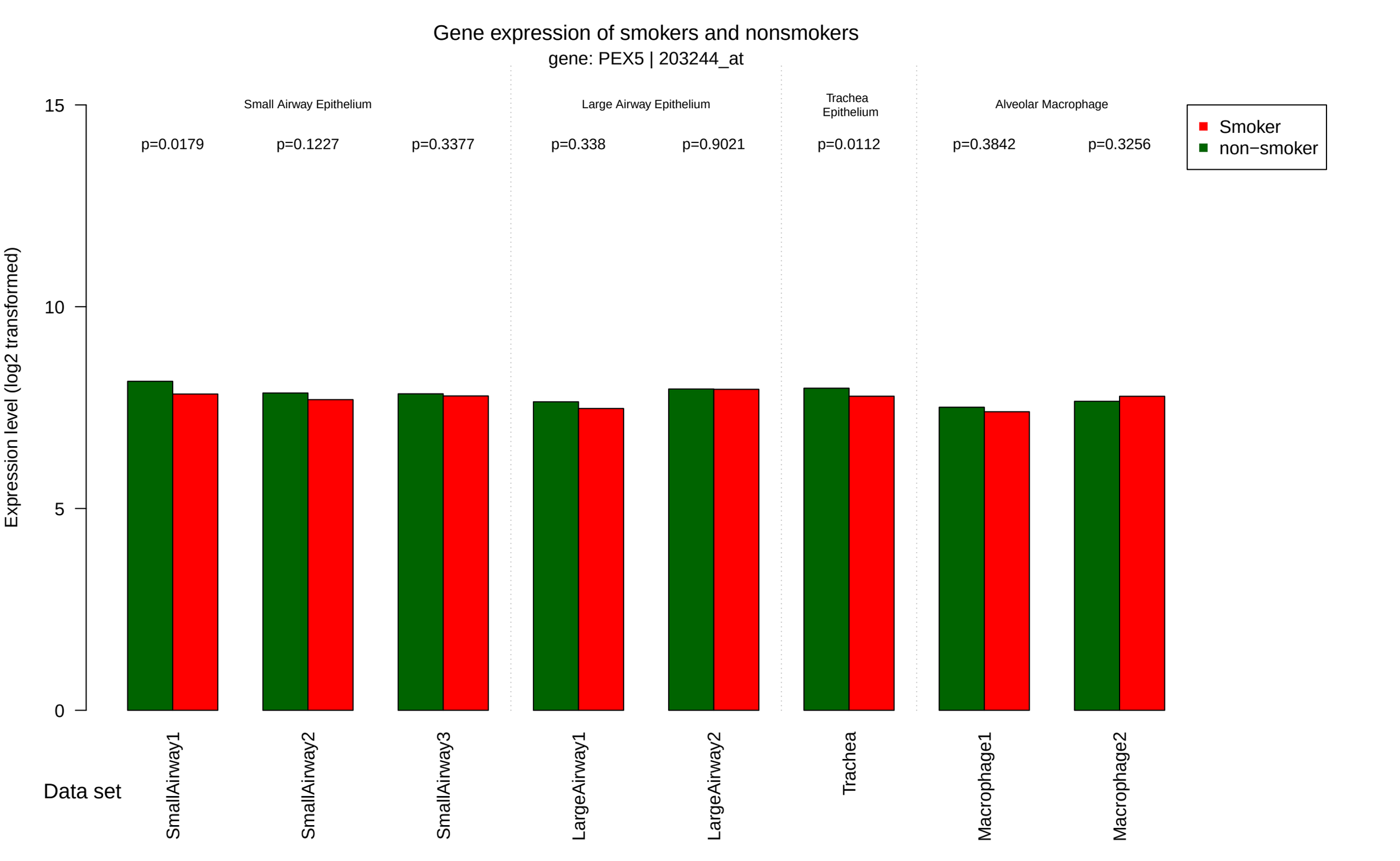
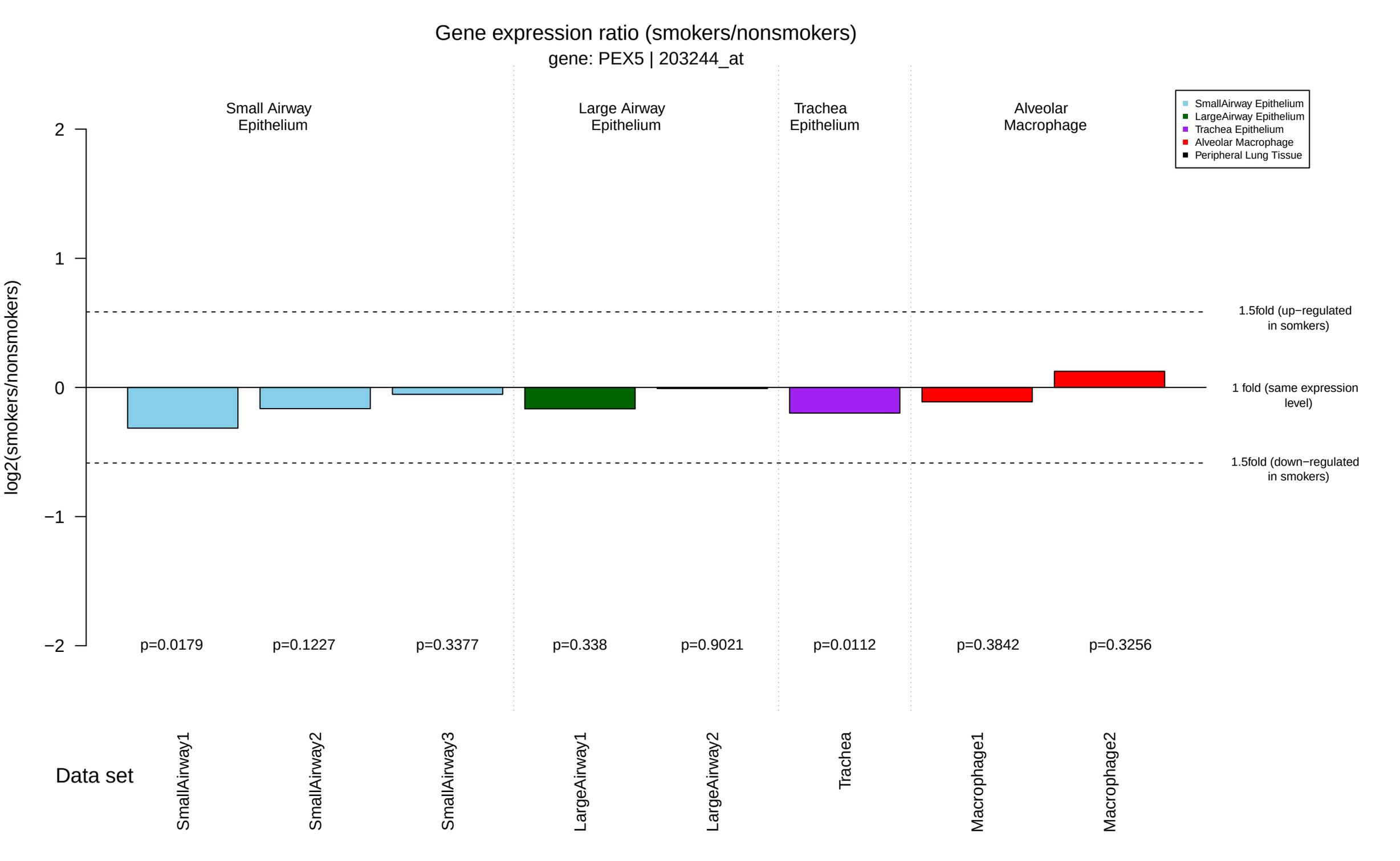
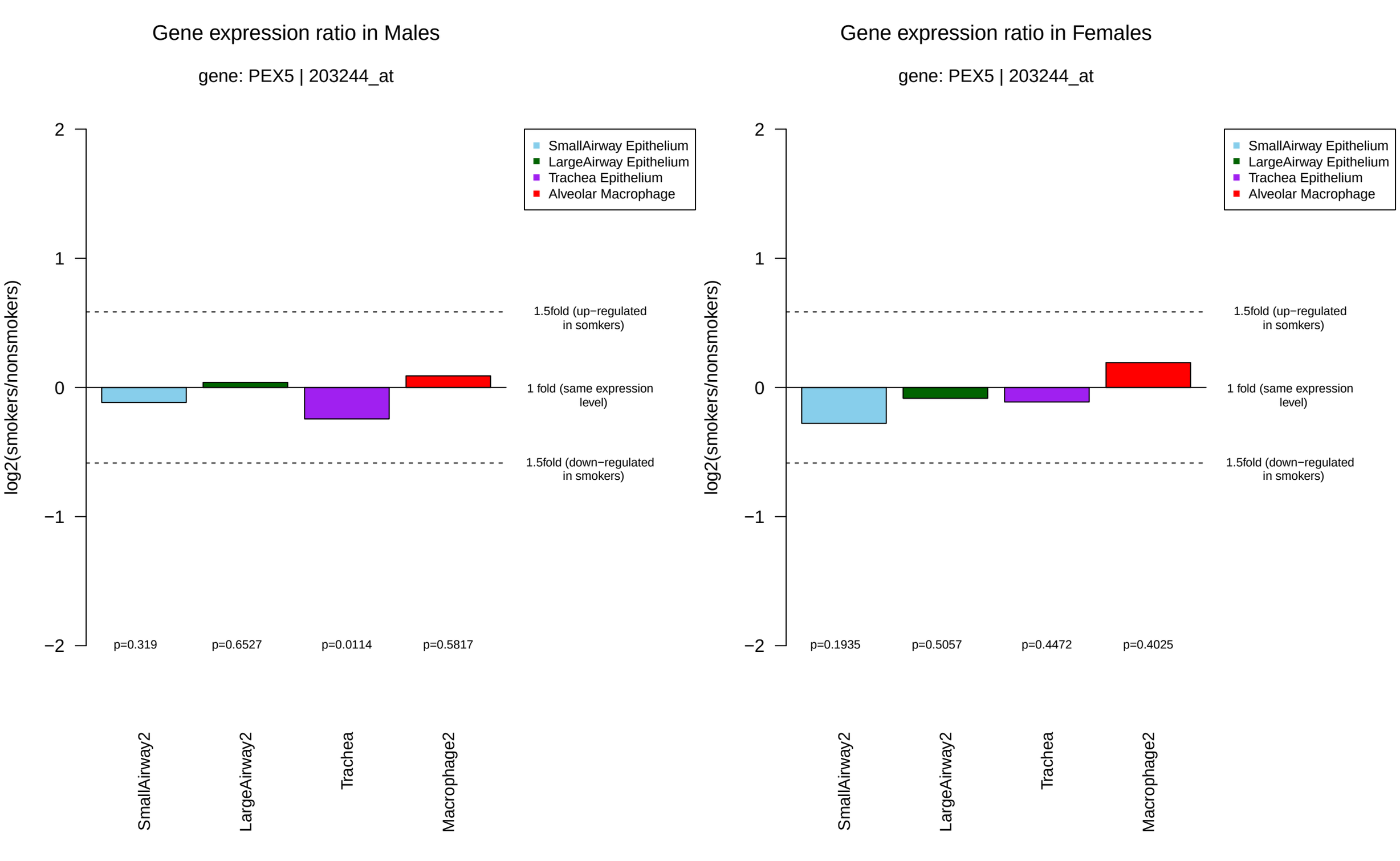
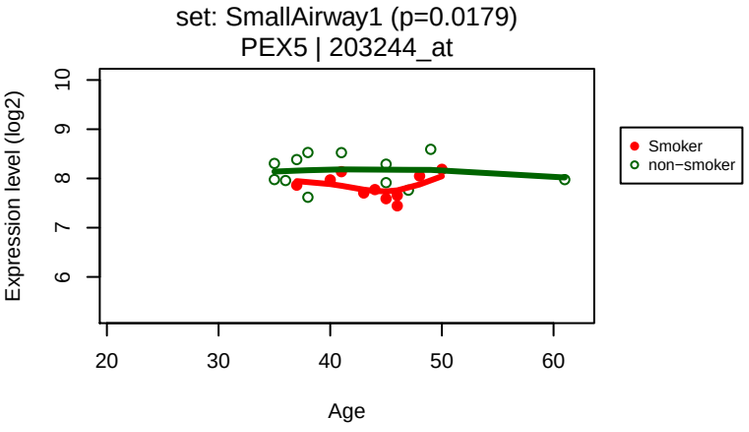
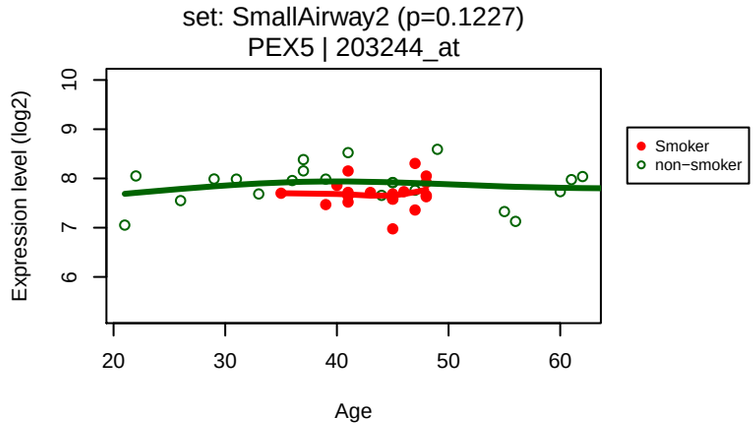
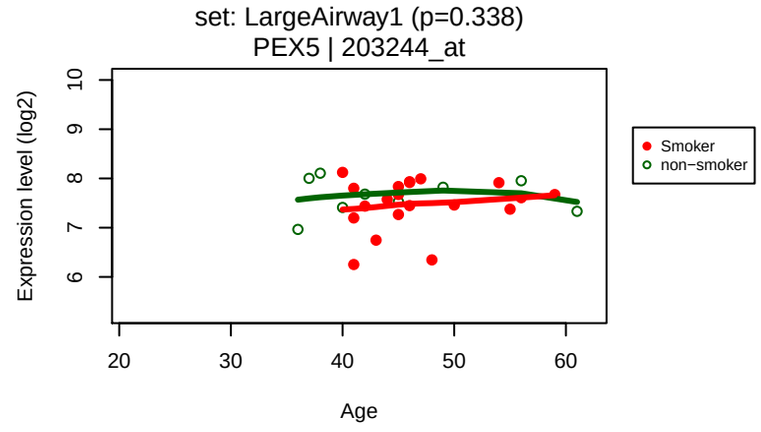

Gene expression of smokers and nonsmokers
gene: PEX5 | 203244_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0179
p=0.1227
p=0.3377
p=0.338
p=0.9021
p=0.0112
p=0.3842
p=0.3256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

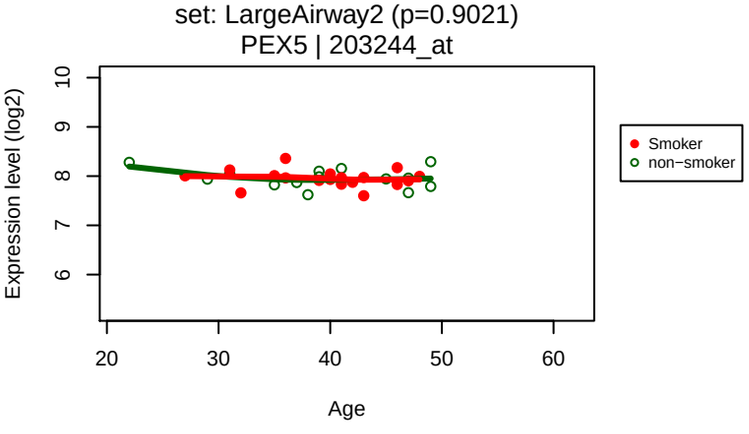

Gene expression ratio (smokers/nonsmokers)
gene: PEX5 | 203244_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0179
p=0.1227
p=0.3377
p=0.338
p=0.9021
p=0.0112
p=0.3842
p=0.3256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

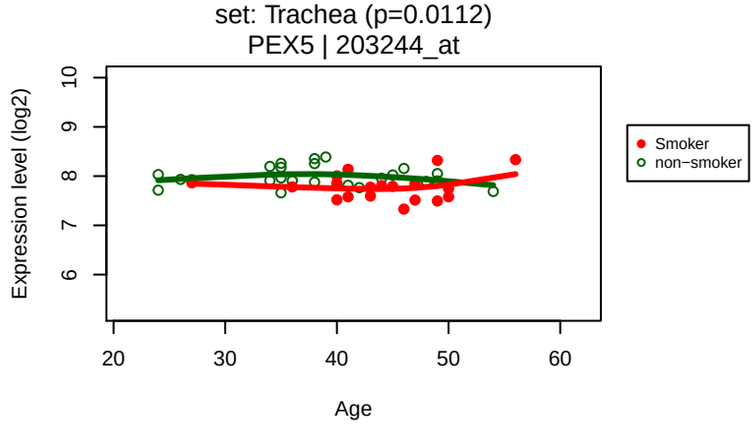

Gene expression ratio in Males
gene: PEX5 | 203244_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.319
p=0.6527
p=0.0114
p=0.5817
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEX5 | 203244_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1935
p=0.5057
p=0.4472
p=0.4025
SmallAirway2
LargeAirway2
Trachea
Macrophage2

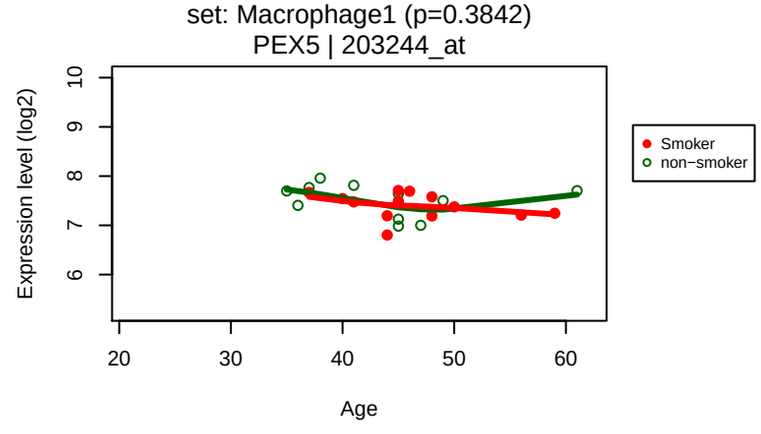

set: SmallAirway1 (p=0.0179)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker

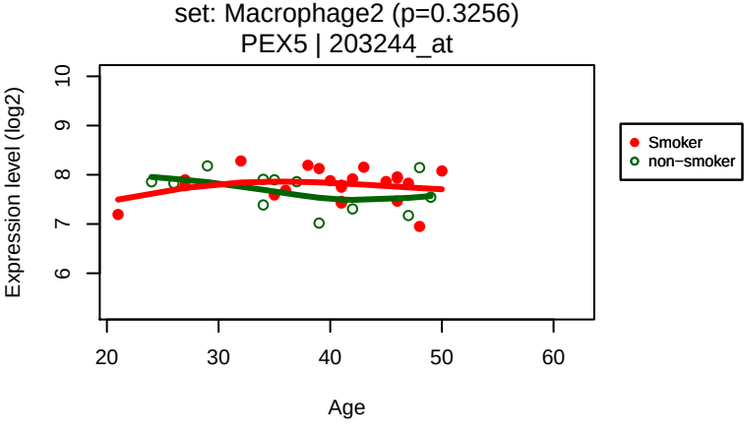

set: SmallAirway2 (p=0.1227)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker

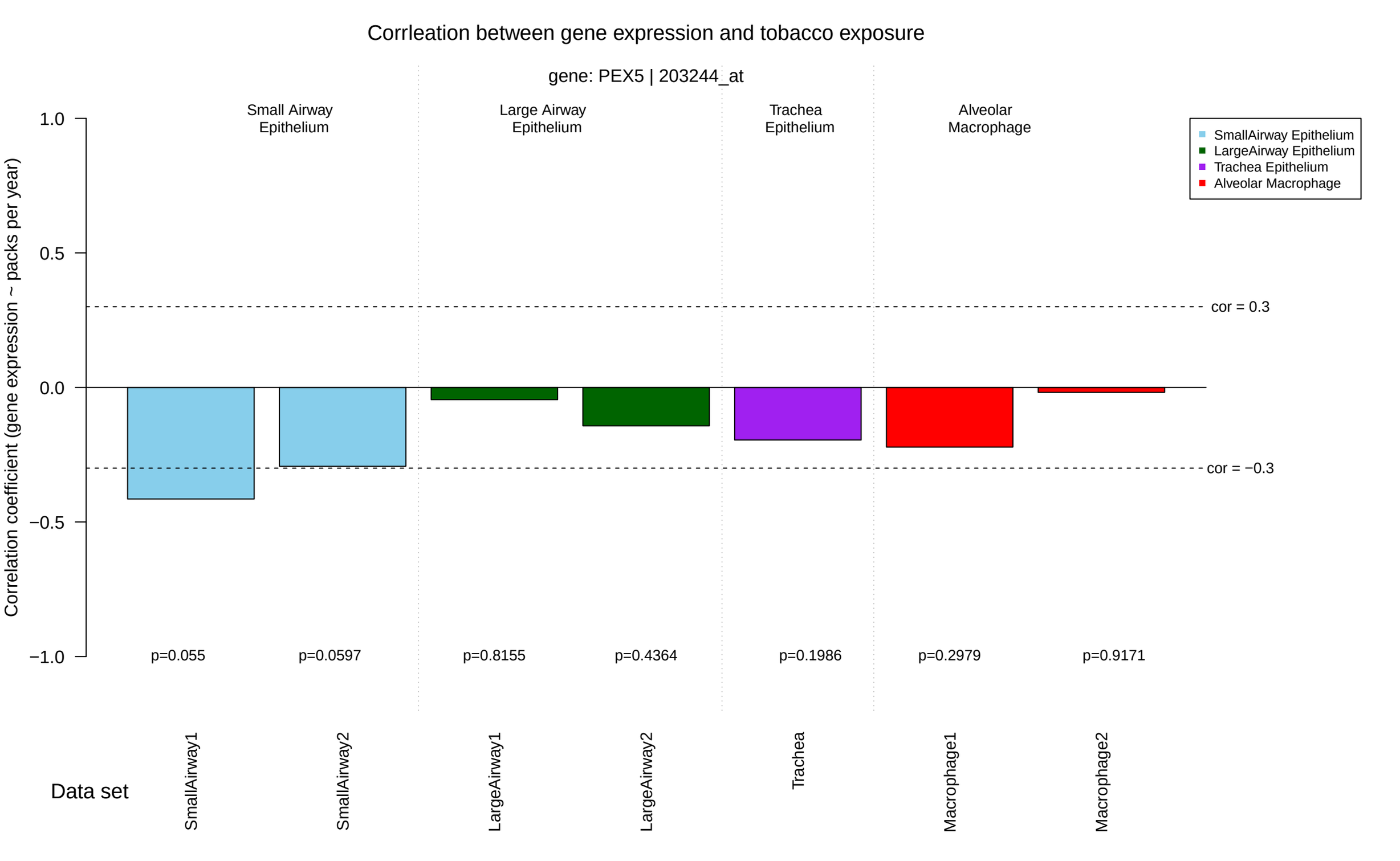

set: LargeAirway1 (p=0.338)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9021)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0112)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3842)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3256)
PEX5 | 203244_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEX5 | 203244_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.055
p=0.0597
p=0.8155
p=0.4364
p=0.1986
p=0.2979
p=0.9171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2